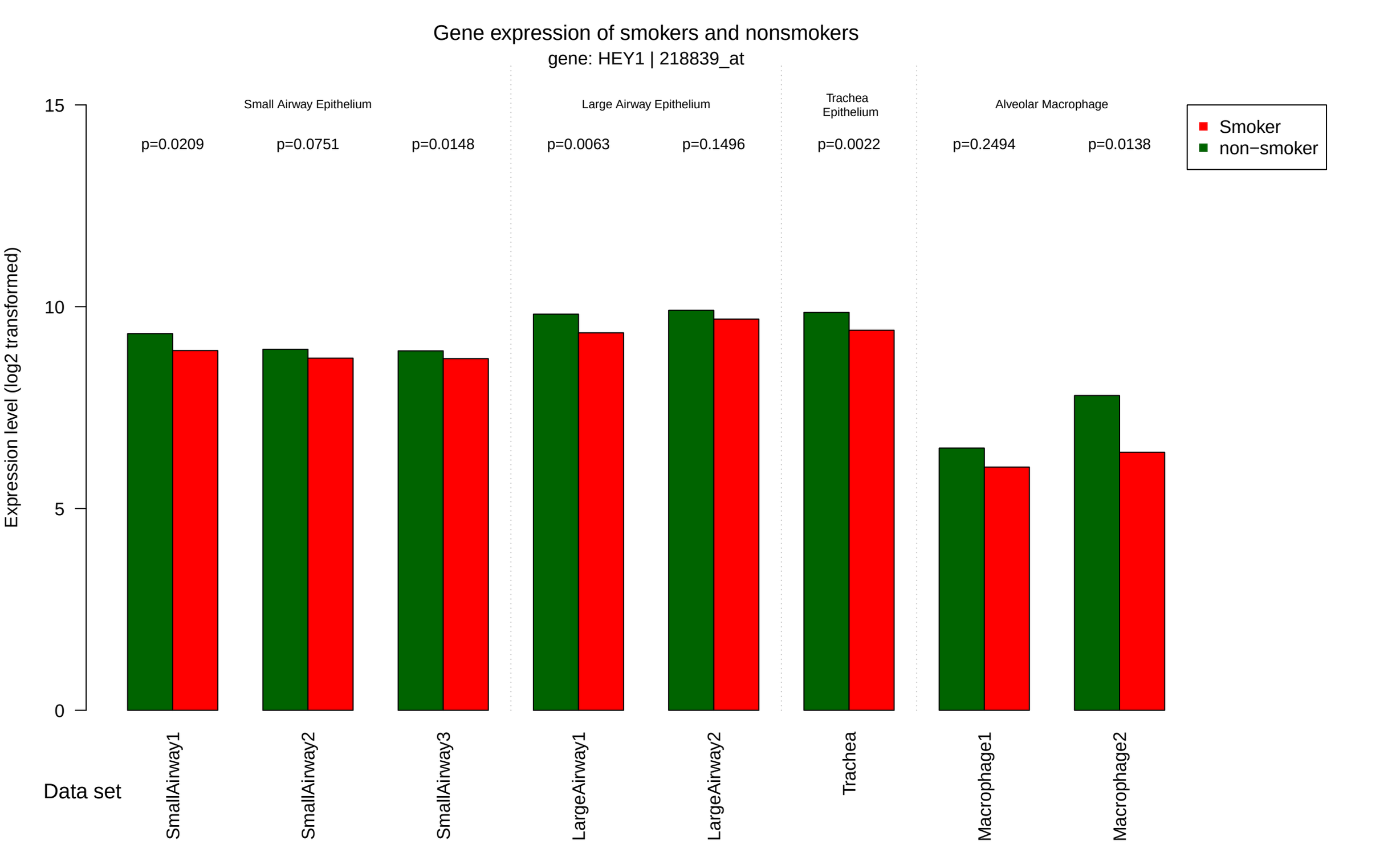
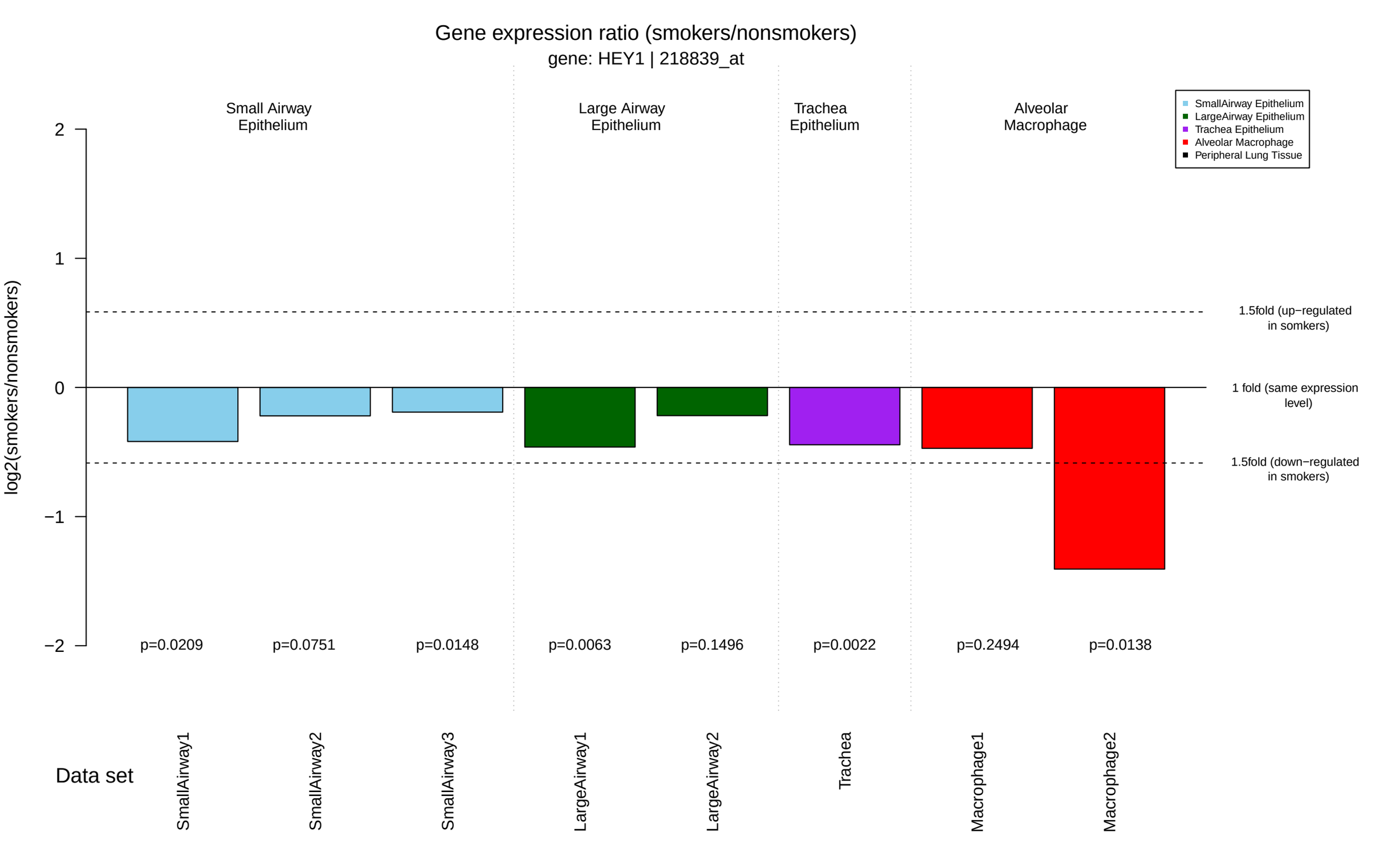
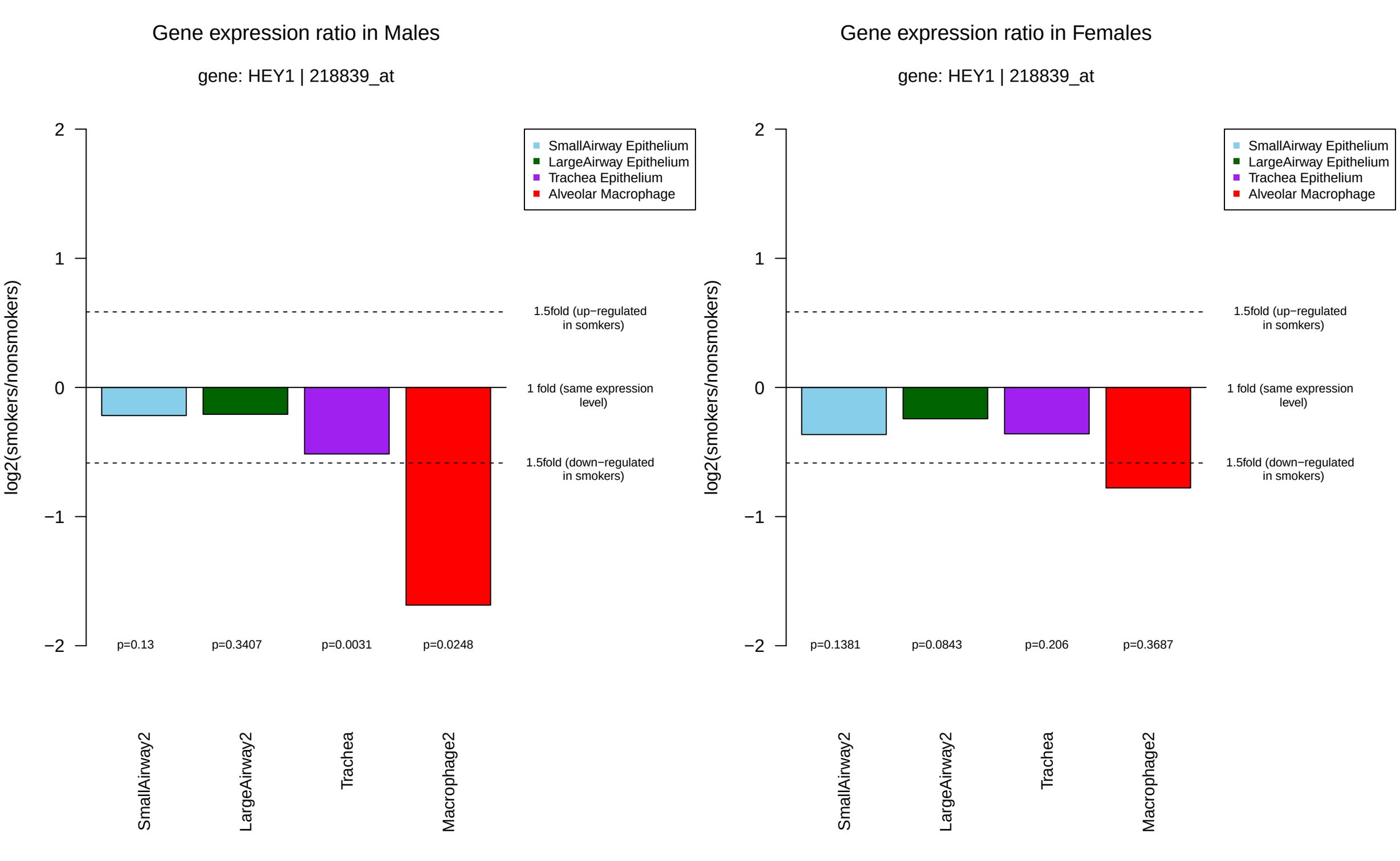
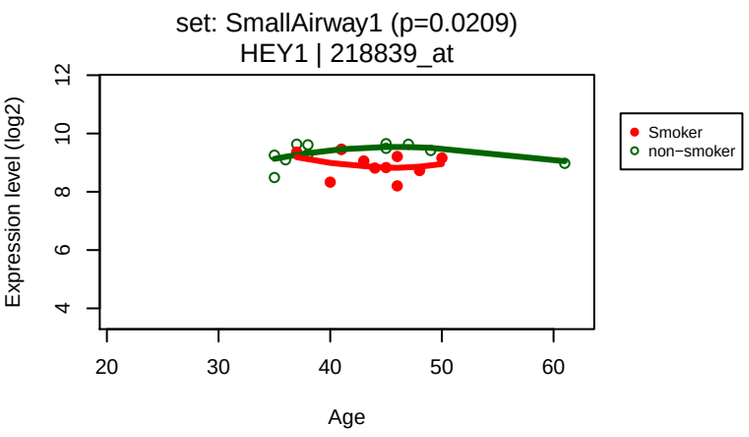
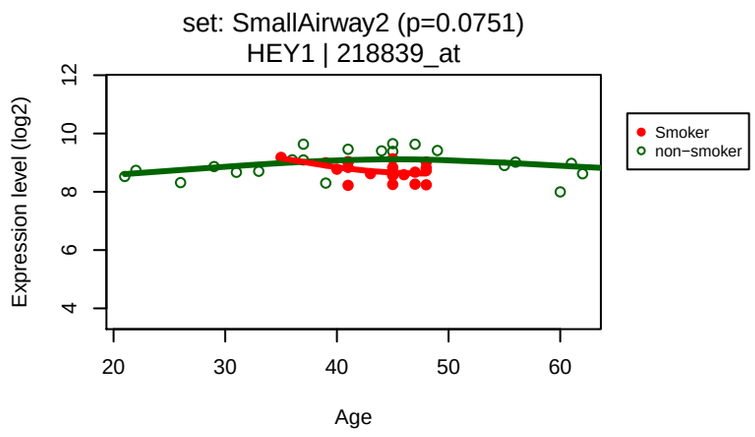
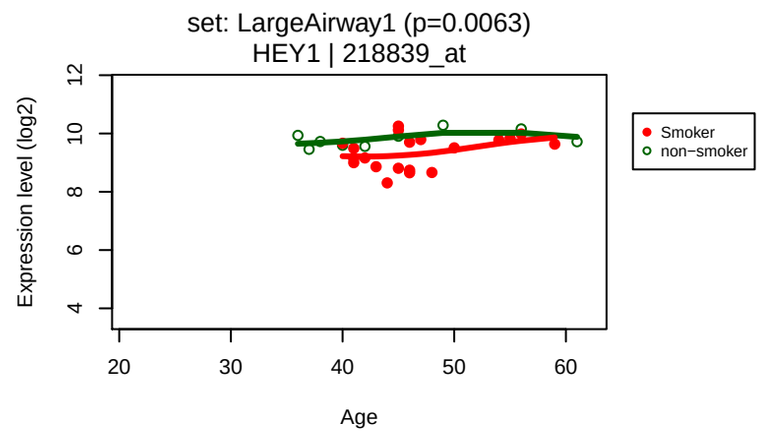

Gene expression of smokers and nonsmokers
gene: HEY1 | 218839_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0209
p=0.0751
p=0.0148
p=0.0063
p=0.1496
p=0.0022
p=0.2494
p=0.0138
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

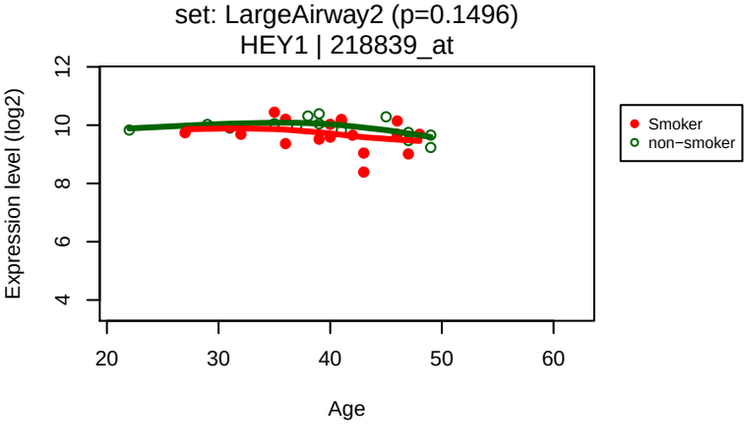

Gene expression ratio (smokers/nonsmokers)
gene: HEY1 | 218839_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0209
p=0.0751
p=0.0148
p=0.0063
p=0.1496
p=0.0022
p=0.2494
p=0.0138
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

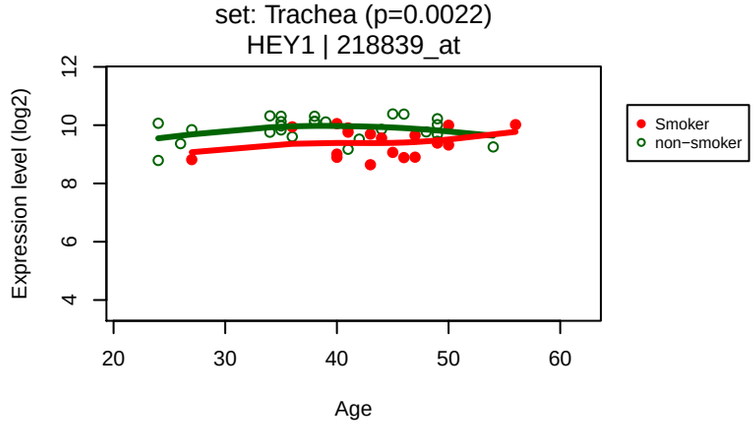

Gene expression ratio in Males
gene: HEY1 | 218839_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.13
p=0.3407
p=0.0031
p=0.0248
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HEY1 | 218839_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1381
p=0.0843
p=0.206
p=0.3687
SmallAirway2
LargeAirway2
Trachea
Macrophage2

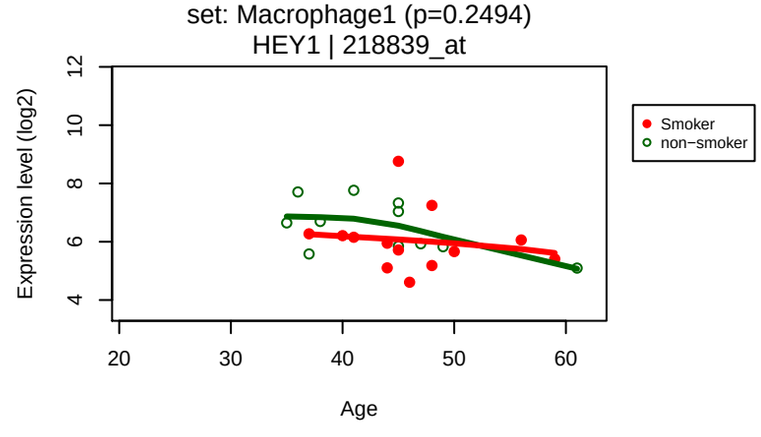

set: SmallAirway1 (p=0.0209)
HEY1 | 218839_at
Expression level (log2)
Age
Smoker
non−smoker

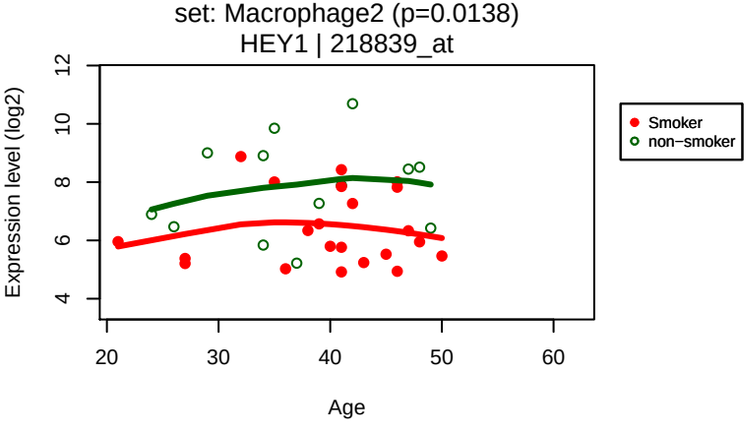

set: SmallAirway2 (p=0.0751)
HEY1 | 218839_at
Expression level (log2)
Age
Smoker
non−smoker

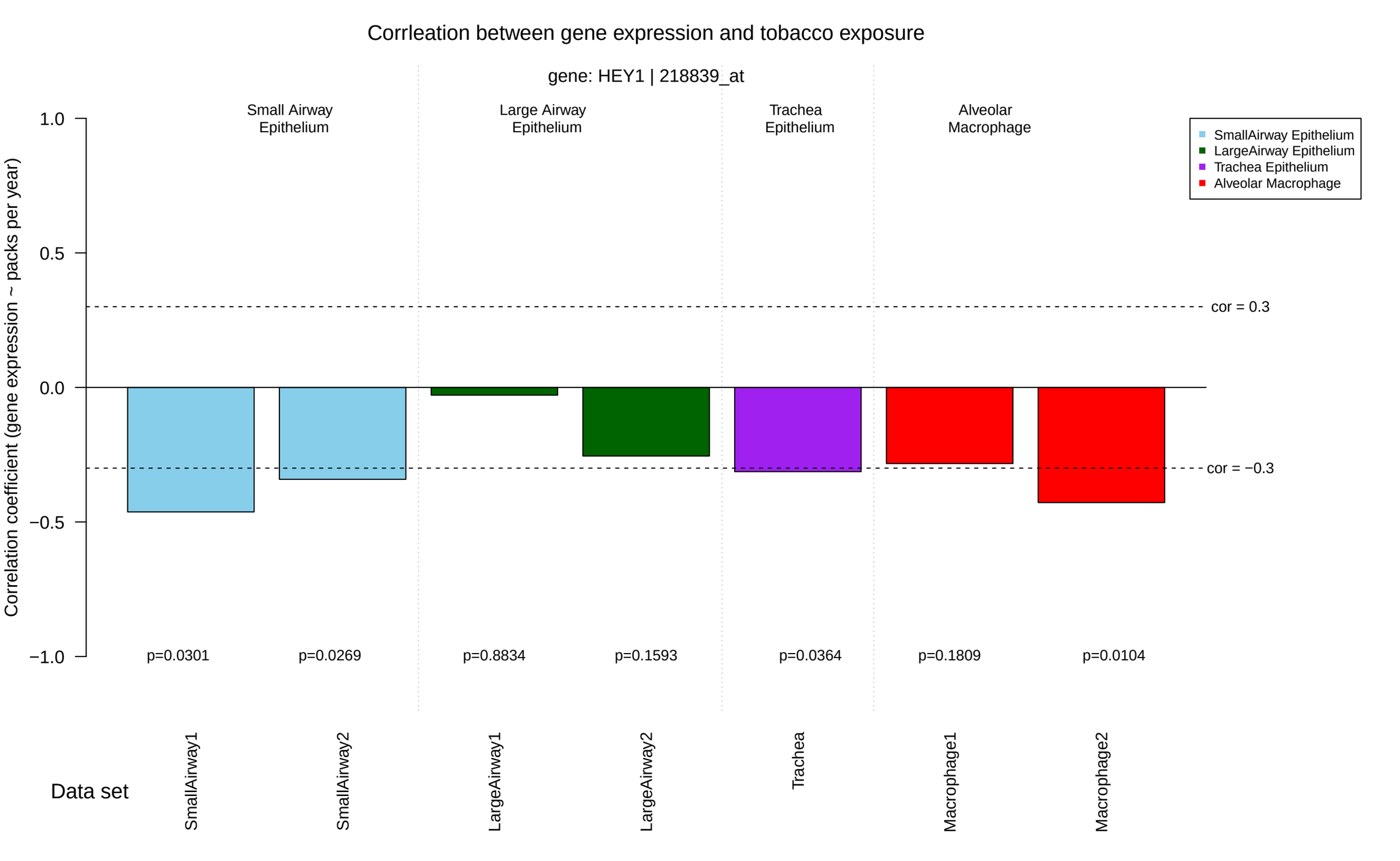

set: LargeAirway1 (p=0.0063)
HEY1 | 218839_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1496)
HEY1 | 218839_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0022)
HEY1 | 218839_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2494)
HEY1 | 218839_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0138)
HEY1 | 218839_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HEY1 | 218839_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0301
p=0.0269
p=0.8834
p=0.1593
p=0.0364
p=0.1809
p=0.0104
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2